Machine Learning Fundamentals

Machine learning is a subset of artificial intelligence that focuses on building systems that learn from data. There are three main types of machine learning: supervised learning, unsupervised learning, and reinforcement learning.

Supervised learning uses labeled training data to learn a mapping from inputs to outputs. Common algorithms include linear regression, logistic regression, decision trees, and neural networks. The model is trained on a dataset where the correct answers are known.

Unsupervised learning finds patterns in data without labeled examples. Clustering algorithms like K-means group similar data points together. Dimensionality reduction techniques like PCA help visualize high-dimensional data.

Reinforcement learning trains agents to make decisions by maximizing cumulative reward. The agent interacts with an environment and learns from the consequences of its actions. Popular algorithms include Q-learning and policy gradient methods.

Deep learning is a subset of machine learning that uses neural networks with multiple layers. Convolutional Neural Networks (CNNs) excel at image recognition. Recurrent Neural Networks (RNNs) and Transformers are used for natural language processing.

Transfer learning allows models trained on one task to be adapted for different tasks. This approach significantly reduces the amount of training data needed and has become the standard approach in modern NLP with models like BERT and GPT.